

Justin Doty, Ph.D.

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BIOGRAPHY

I am a senior application engineer for MathWorks. I received my doctorate in Economics from the University of Iowa with a specialization in econometric modeling and empirical industrial organization.

EXPERTISE

Theoretical and Applied Econometrics - Empirical Industrial Organization - Heterogeneous Treatment Effects - Measurement Error - Quantile Regression - Productivity Estimation - Auctions - Differential Privacy

EDUCATION

The University of Iowa, Iowa City, IA 52242 2016 - 2022

Ph.D. in Economics

Dissertation: *Identification and Estimation of Production Functions with Unobserved Heterogeneity*

Advisor: *Suyong Song*

M.A. in Economics

California State University, Fullerton (CSUF), Fullerton, CA 92834 2011 - 2016

B.A. in Economics, minor in Mathematics

EXPERIENCE

MathWorks Washington D.C, 20233 Feb 2026 - present

Finance Division

Job Title: Senior Application Engineer - Econometric Modeling

- Deliver demonstrations and engage in technical engagements with economists

U.S. Census Bureau Washington D.C, 20233 Aug 2022 - Jan 2026

Center for Enterprise Dissemination - Disclosure Avoidance, Research and Applications Group

Job Title: Research Economist

- Modernized statistical disclosure methodology for protecting data confidentiality
- Developed formally private algorithms for heavy-tailed, business establishment level data
- Developed and tested statistical programs to protect categorical data using local differential privacy
- Developed and tested statistical programs to correct for inferences made on privacy-protected data
- Tested experimental iterations of the Disclosure Avoidance System (DAS)
- Collaborated on research papers that developed formally private algorithms for different data release scenarios
- Reviewed and approved internal working research papers for public release
- Provided subject-matter expertise on statistical disclosure methods for planned external data releases

The University of Iowa, Iowa City, IA 52242 2016 - 2022

Job Title: Graduate Teaching Assistant

- Assisted professors in teaching the following courses: *Ph.D. Econometrics, Statistics for Strategy, Principles of Microeconomics, Principles of Macroeconomics, Global Economics*
- Assisted professors in grading the following courses: *Ph.D. Econometrics, Ph.D. Statistics for Economics, Ph.D. Applied Econometrics, Economic and Business Forecasting, Managerial Economics*
- Assisted economics graduate students in their econometric theory questions and assignments
- Provided supplemental instruction in econometric theory to economics graduate students
- Assisted both undergraduate and graduate students in learning statistical programming languages
- Held office hours and review sessions for students outside of discussion sessions

PUBLISHED PAPERS

Finley, Caruso, Doty, Machanavajjhala, Meyer, Pujol, Sexton, and Terner. (2026). “Slowly Scaling Per-Record Differential Privacy”. *Journal of Privacy and Confidentiality* 16 (1). <https://doi.org/10.29012/jpc.992>.

WORKING PAPERS

“A Dynamic Framework for Identification and Estimation of Nonseparable Production Functions”
“Estimating Quantile Production Functions: A Control Function Approach” (*with Suyong Song*)

WORKS IN PROGRESS

“Multinomial Randomized Response for Data Sharing (*with Caleb Floyd and Rolando Rodriguez*)”
“Identification of Gross-Output Production Functions with Mismeasured Inputs”
“Estimating Industry Competitor Network Effects on Firm-Level Productivity”

AWARDS AND FELLOWSHIPS

Ballard and Seashore Dissertation Fellowship, University of Iowa	2021
Summer Fellowship, University of Iowa	2020
Post-Comprehensive Research Fellowship, University of Iowa	2020
GSS/Graduate College Presentation Travel Funds, University of Iowa	2019
Stewart-Long Graduate Study Award, CSUF	2015
Outstanding Student Scholarly and Creative Activities Award, CSUF	2015

INVITED SEMINARS AND PRESENTATIONS

American Economic Association (AEA/ASSA) Annual Meeting	January 2025
North America Summer Meeting of the Econometric Society (NASMES)	June 2022
European Winter Meeting of the Econometric Society (EWMES)	December 2021
Southern Economic Association (SEA)	November 2021
Joint Statistical Meetings (JSM)	August 2020
Midwest Econometrics Group (MEG)	October 2019

SKILLS

Programming & Scripting: R (tidyverse, data.table, quantreg, parallel, rstan) Python (pandas, numpy, matplotlib, scikit-learn, pytorch, pyspark), Matlab, Vim
Tools & Environments: Git, GitHub, Jupyter Notebooks, AWS EC2, Gurobi, Stan
Concepts: Linear-Logistic-Multinomial Regression, Nonlinear Models, Kernel Density Estimators, Monte Carlo Simulations, Large-Sample Theory, Machine Learning Algorithms, Time Series Forecasting, Small Area Estimation, Differential Privacy and Formal Privacy Variants, Database Reconstruction and Membership Inference Attacks

CERTIFICATIONS AND TRAININGS

Machine Learning Specialist - U.S. Census Data Science Training Program	2025
Global Leadership - Student Leadership Institute, CSUF	2016

JOURNAL REFEREE EXPERIENCE

Journal of Empirical Economics

REFERENCES

References available upon request